

Did Post-Training Make Models Loss-Averse?

Structural Estimation of Reference-Dependent Preferences in Large Language Models

J. Blakeslee

Pardee RAND Graduate School

Summer OJT Practicum (Course 1012)

Faculty: Todd Richmond

June 29 – September 4, 2026

Abstract

This project treats large language models as economic agents and recovers their preference primitives from their choices. I elicit thousands of binary risky choices under controlled experimental variation, then estimate structural discrete-choice models on the resulting data: first a risk-aversion coefficient from lottery menus, then loss aversion and an endogenous reference point from systematically framed gains and losses, in the tradition of Kahneman and Tversky and Kőszegi and Rabin. Recent work has shown, descriptively, that instruction-tuned models display weak and unstable loss aversion. This project asks the question that work leaves open: *where do these preferences come from?* By running the identical elicitation on matched base and instruction-tuned versions of open-weight models, I test whether post-training (instruction tuning and RLHF) instills or amplifies reference-dependent behavior. The design is built around the measurement problems now documented in this literature: training-data contamination, position bias, prompt-template sensitivity, and the questionable mapping from sampling stochasticity to choice noise. The deliverables are a reproducible elicitation-to-estimation pipeline, a documented dataset of model choices, and a short working paper reporting estimated parameters, confidence intervals, and the base-versus-instruct contrast.

1 The Research Questions

The project answers three questions, in ascending order of novelty:

- Q1. Do language models exhibit stable, structured risk preferences?** When a model makes risky choices, can its behavior be summarized by interpretable primitives (a risk-aversion coefficient, a loss-aversion parameter, a reference point) recovered by maximum likelihood at the level of individual choices?
- Q2. Are those preferences consistent under reframing?** The same underlying choice, framed once as a gain and once as a loss, should produce the same decision from a rational agent. Humans famously violate this. Does the model, and by how much in estimated-parameter terms?

Q3. Did post-training put the preferences there? (Headline.) Running the identical elicitation on matched base and instruction-tuned versions of the same open-weight model attributes any estimated loss aversion or frame sensitivity to post-training. If the base model is roughly frame-invariant and the instruct model is loss-averse, alignment training transferred a human cognitive bias into the model along with human values. Either direction of the result is informative.

Each question resolves to estimated numbers with standard errors and a sharp hypothesis test, and a clean null is itself a reportable finding.

2 Why These Questions Matter

2.1 Models are becoming economic agents

Language models increasingly make allocative decisions in deployment: pricing, resource allocation, negotiation, and risk judgments inside agentic systems. The behavior of an agent that acts on the world is a governance object. Revealed-preference estimation is the discipline economics built for exactly this purpose, and it has only begun to be pointed at models.

2.2 Behavioral consistency is an alignment property

A model whose choices reverse when a problem is reframed is, in a precise sense, manipulable. The same susceptibility that makes a human buy the extended warranty makes a model exploitable by an adversarial prompt or an interface designed to steer it. Measuring frame invariance in estimated-parameter terms turns an anecdote into a robustness result stated in the language of choice theory.

2.3 Attribution to post-training is the open causal question

Itzhak, Stanovsky, Rosenfeld, and Belinkov (2024) showed qualitatively that instruction tuning introduces or amplifies cognitive biases (decoy, certainty, belief bias) relative to base models. No one has made that attribution *structurally*, for risk and loss preferences, with parameters estimated from choices. That is the causal question alignment teams care about: what kind of economic agent does post-training actually produce? This project’s base-versus-instruct arm answers it with an estimated λ on each side of the training intervention.

2.4 The contribution is structural, which buys counterfactuals

Showing that a model behaves oddly in one experiment is a description. Recovering the primitives behind its behavior lets me predict its choices in environments I never ran. The structural step is what makes the result generalize, and it is the step most adjacent work omits.

2.5 Positioning in the literature (as of mid-2026)

The descriptive generation of this literature is complete: Horton (2023) and Binz and Schulz (2023) established that models qualitatively reproduce classic behavioral results; Mei et al. (2024) showed broad distributional similarity to humans; Chen, Liu, Shan, and Zhong (2023) found GPT-4 highly

consistent with utility maximization by revealed-preference (GARP) tests; and Bini, Cong, Huang, and Jin (2026) survey the space of LLM biases at econ-venue scale, descriptively.

A structural generation is now emerging, and this project extends its two incumbents. Liu, Yang, and Tam (2025) fit prospect-theory parameters to LLM certainty equivalents by curve-fitting and report preferences unstable across prompts and temperatures. A 2025 cumulative-prospect-theory study (arXiv:2508.08992) estimates full CPT parameters on instruct models and finds loss aversion far below human benchmarks and unstable (λ ranging from near 0 to ≈ 1.9 against a human ≈ 2.6). Mazeika et al. (2025) fit Thurstonian random-utility models to LLM binary choices, though over general outcomes rather than lotteries. Related calibration work (Yee and Koh 2026) likewise documents attenuated loss aversion at baseline in finance settings.

What no paper combines, and where this project sits: (i) choice-level maximum-likelihood estimation on binary lottery choices, rather than curve-fitting to derived certainty equivalents; (ii) a Kőszegi-Rabin loss-aversion parameter *and endogenous reference point* identified from explicit gain/loss frame manipulation; (iii) a formal frame-invariance test; and (iv) a matched base-versus-instruct contrast that attributes the estimated parameters to post-training. Given the incumbents' findings, my prior is that instruct models will show attenuated, unstable loss aversion; the contribution is not re-documenting that fact. The contribution is the attribution design and the measurement-honest estimation behind it.

3 Methodology

The project runs in three phases. The first is a deliberately easy warm-up that de-risks the entire pipeline. The second adds the framing manipulation. The third carries the identical instruments to matched base and instruct models.

3.1 Phase 1 (warm-up): risk preferences from lottery menus

Following the multiple-price-list tradition (Holt and Laury 2002; Andersen et al. 2008) in structure but never in wording (see §3.4), I present the model with paired gambles across a procedurally generated grid of stakes and probabilities. Each trial is an independent call with no memory of prior choices. I model utility as constant relative risk aversion,

$$u(x) = \frac{x^{1-r}}{1-r},$$

with a logit (Fechner) choice error, so the probability of choosing lottery B over option A is

$$\Pr(B) = \frac{1}{1 + \exp(-(EU_B - EU_A)/\mu)},$$

and estimate r and μ by maximum likelihood on the individual choices. This is the canonical structural exercise, checkable against any first-year problem set, which is why it is the right place to debug elicitation before the stakes rise.

3.2 Phase 2 (main study): reference dependence and loss aversion

The same lottery is presented twice: once framed as a gain from a low reference point and once as a loss from a high reference point (“you start with \$117; option B risks losing all of it”). The

underlying payoffs are identical; only the frame moves. I extend the utility specification to carry a reference point ρ and a loss-aversion parameter λ , in the spirit of Tversky and Kahneman (1991) and Kőszegi and Rabin (2006):

$$v(x \mid \rho) = \begin{cases} (x - \rho)^\alpha, & x \geq \rho \\ -\lambda (\rho - x)^\beta, & x < \rho, \end{cases}$$

and estimate λ , the reference point’s sensitivity to the frame, and the curvature terms by maximum likelihood. The headline tests are sharp hypotheses:

- $H_0 : \lambda = 1$ (no loss aversion; the rational benchmark).
- H_0 : the estimated reference point does not move with the frame (frame invariance).

Because recent work suggests frontier instruct models may show weak framing effects, a null here is a plausible *true* result. The design therefore includes a positive control: a manipulation with a documented large effect, so that a weak framing estimate is interpretable as a finding about the model rather than a failure of the instrument.

3.3 Phase 3 (attribution): base versus instruct

The identical instruments run on matched open-weight pairs (e.g., Llama-3.1-8B base and instruct, or a Qwen pair), giving estimated (r, λ, ρ) on each side of the post-training intervention. Base models do not follow instructions, so this arm elicits choices through few-shot completion formatting and reads choice probabilities directly from token log-probabilities rather than sampled letters. Logprob elicitation is also a methodological upgrade in its own right: it measures $\Pr(\text{choose } B)$, the exact quantity the logit likelihood is built on, in a single call.

3.4 Design commitments: the measurement model taken seriously

The validity threats in this literature are now documented, and the design treats them as first-order rather than as robustness afterthoughts.

- **Contamination-safe instruments.** Canonical vignettes (Holt-Laury menus, Ellsberg urns, Asian-disease framing) are almost certainly in training corpora, so a verbatim replication risks recovering a memorized textbook answer. Every lottery in this study is procedurally generated with randomized, non-round payoffs and probabilities inside novel cover stories, and an explicit contamination probe documents that the instruments are not being recited.
- **Counterbalanced position and labels.** First-option bias is among the largest documented LLM artifacts. Each choice is presented in both orders with safe/risky roles swapped; the residual position effect is estimated as a nuisance parameter rather than assumed away.
- **Prompt template as an experimental factor.** Elicited preferences are known to shift under paraphrase. Each lottery is drawn under five or more semantically equivalent templates, with the template entering the choice model as a random effect and between-template variance reported alongside the point estimates.
- **Honest treatment of choice stochasticity.** Sampling at temperature is not a validated random-utility error process. Where log-probabilities are available (open models), choice

probabilities are read directly and debiased for token priors; for closed models, sampled frequencies are collected at two temperatures and the sensitivity of the error structure to the sampling knob is reported.

- **Reproducibility and framing of inference.** Every run pins a dated model snapshot logged in the released dataset, and a calibration battery re-runs on a second snapshot to bound drift. Recovered parameters are presented as properties of the model’s conditional text distribution under this protocol, following emerging best practice for LLM-as-subject studies, with an internal-consistency (transitivity) check as validation.

A full Phase 2 dataset is roughly $40 \text{ gambles} \times 2 \text{ frames} \times 5 \text{ templates} \times \text{counterbalanced orders} \times \text{repetitions}$, on the order of a few thousand calls per model, which costs a few dollars and runs in about an hour.

4 What I Will Learn

This project is chosen as a skill-building artifact, one deliberate notch past my current toolkit on two axes at once.

- **Model elicitation and experimental design on AI systems.** Designing controlled experiments for a model rather than a human: contamination-proofing, counterbalancing, template randomization, forced structured output, logprob elicitation, and the treatment of sampling stochasticity. This is the skill that makes the work legible to a lab evaluations team.
- **Structural estimation of preference primitives.** My empirical training is reduced-form and causal (difference-in-differences, synthetic control, instrumental variables). This project requires likelihood-based recovery of behavioral primitives, and the design commitments push the econometrics further: nuisance parameters, template random effects (a mixed-logit structure), and variance decomposition. That is a deeper structural education than a plain logit fit.
- **The bridge between behavioral economics and AI evaluation.** Stating an alignment-relevant property (frame invariance, manipulability, post-training attribution) as a hypothesis test on estimated parameters.
- **Reproducible research engineering.** A public pipeline from elicitation to dataset to estimation to writeup, re-runnable by a reviewer against pinned model snapshots.

5 Timeline

Weeks	Milestone
1–2	Harness with design commitments built in (procedural instruments, counterbalancing, templates); pilot data from Claude; contamination probe run.
3	Phase 1 CRRA estimate on Claude; validation checks pass; warm-up result locked.
4–5	Phase 2 design: frames, positive control, pre-registered analysis plan; Phase 2 run.
6	Phase 3: few-shot/logprob elicitation on a matched base-instruct open-model pair.
7–8	Estimation: mixed-logit with template random effects; λ and reference point; the two sharp hypothesis tests; the base-versus-instruct contrast.
9	Robustness: second snapshot, two temperatures, reason-then-answer arm as time allows; draft writeup.
10	Final briefing to the cohort; public reproducible repository and short paper.

A presentable result exists by week three regardless of how the later phases unfold, so the finished deliverable is secured early and the remaining weeks add ambition rather than risk.

6 Deliverables

- A reproducible code repository: elicitation harness plus estimation, runnable end to end against pinned model snapshots.
- A documented dataset of model choices under every condition, with snapshot IDs, seeds, and discard diagnostics.
- A short working paper reporting estimated parameters, confidence intervals, the frame-invariance verdict, and the base-versus-instruct attribution.
- The final sprint briefing, with a public-facing explainer of the headline finding.

7 Fit with the Sprint

The practicum asks for applications work that informs implications work. This project is that loop in miniature: the application is a working model-evaluation pipeline I build and run; the implication is a governance-relevant claim about whether automated economic agents carry stable, manipulable preferences and whether alignment training is the source. The technical build produces the evidence, and the evidence produces the argument. A lab evaluations or economics team reads the result as rigorous model elicitation; a policy reader reads it as a concrete statement about how an AI agent behaves under pressure, and why.

References

- [1] Andersen, S., Harrison, G. W., Lau, M. I., and Rutström, E. E. (2008). Eliciting Risk and Time Preferences. *Econometrica*, 76(3), 583–618.
- [2] Bini, S., Cong, L. W., Huang, D., and Jin, L. (2026). Behavioral Economics of AI: LLM Biases and Corrections. *NBER Working Paper* 34745.

- [3] Binz, M., and Schulz, E. (2023). Using Cognitive Psychology to Understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6).
- [4] Chen, Y., Liu, T. X., Shan, Y., and Zhong, S. (2023). The Emergence of Economic Rationality of GPT. *Proceedings of the National Academy of Sciences*, 120(51).
- [5] Holt, C. A., and Laury, S. K. (2002). Risk Aversion and Incentive Effects. *American Economic Review*, 92(5), 1644–1655.
- [6] Horton, J. J. (2023). Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus? *NBER Working Paper* 31122.
- [7] Itzhak, I., Stanovsky, G., Rosenfeld, N., and Belinkov, Y. (2024). Instructed to Bias: Instruction-Tuned Language Models Exhibit Emergent Cognitive Bias. *Transactions of the Association for Computational Linguistics*, 12.
- [8] Kahneman, D., and Tversky, A. (1979). Prospect Theory: An Analysis of Decision under Risk. *Econometrica*, 47(2), 263–291.
- [9] Köszegi, B., and Rabin, M. (2006). A Model of Reference-Dependent Preferences. *Quarterly Journal of Economics*, 121(4), 1133–1165.
- [10] Liu, J., Yang, Y., and Tam, K. Y. (2025). Evaluating and Aligning Human Economic Risk Preferences in LLMs. *Proceedings of EMNLP 2025*; arXiv:2503.06646.
- [11] Mazeika, M., et al. (2025). Utility Engineering: Analyzing and Controlling Emergent Value Systems in AIs. arXiv:2502.08640.
- [12] Mei, Q., Xie, Y., Yuan, W., and Jackson, M. O. (2024). A Turing Test of Whether AI Chatbots Are Behaviorally Similar to Humans. *Proceedings of the National Academy of Sciences*, 121(9).
- [13] Train, K. E. (2009). *Discrete Choice Methods with Simulation* (2nd ed.). Cambridge University Press.
- [14] Tversky, A., and Kahneman, D. (1991). Loss Aversion in Riskless Choice: A Reference-Dependent Model. *Quarterly Journal of Economics*, 106(4), 1039–1061.
- [15] Yee, B., and Koh, P. (2026). Large Language Models as Calibrated Measurement Instruments for Behavioral Parameters. arXiv:2602.01022.

Additional methodological sources cited in the design (author lists pending verification for some preprints): CPT estimation on LLMs (arXiv:2508.08992); sampling stochasticity (arXiv:2604.06543); position bias (Zheng et al., ICLR 2024; Pezeshkpour and Hruschka 2024); format-restriction effects (Tam et al., EMNLP 2024); machine-psychology contamination standards (Hagendorff et al., arXiv:2303.13988); validity conditions for LLM-as-subject inference (arXiv:2602.15785). Full list in the project’s literature notes.